

REPORT
OF
DEVELOPMENT, DEPLOYMENT AND MAINTENANCE OF
THUMBBYX WEB APPLICATION USING MERN STACK AND
AWS

Submitted

To

Ms. Surabhi Purwar

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

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Batch: 2023–2027

CANDIDATE'S DECLARATION

I, Shakshi Kumari Maurya do hereby declare that the internship report titled **“Development, Deployment and Maintenance of ThumbbyX Web Application Using MERN Stack and AWS”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I express my gratitude to my parents for their blessings.

Signature

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INTERNSHIP COMPLETION CERTIFICATE

Employee ID: TBX-SDE-0507		Certificate No. TBX-IC-2026-0001
CERTIFICATE OF COMPLETION		
This Certificate is given to : Shakshi Kumari Maurya		
for successfully completing the internship as a Software Developer Intern in the Technology & Product Department at ThumbbyX Global Private Limited from 03 June 2026 to 02 September 2026. During this period, she demonstrated excellent technical abilities, dedication, professionalism, and a strong willingness to learn. We appreciate her valuable contributions to the team and wish her continued success in all future endeavors		
Certification Date: 04 September 2026		
ThumbbyX Global Pvt. Ltd.  Director ADITYA KUMAR SINGH Founder and Director	 SONALI SINGH HR Department	

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
	Cover Page	
1.	Candidate's Declaration	i
2.	Acknowledgment	ii
3.	Internship Completion Certificate	
4.	Project Description	
	4.1 Introduction	1
	4.2 Organization Profile	2
	4.3 Problem Statement	3
	4.4 Project Objectives	4
	4.5 Scope of the Project	5
	4.6 Technologies and Tools Used	5-6
	4.7 System Architecture	6-7
	4.8 Methodology	8-9
	4.9 Expected Outcomes	10-11
	4.10 Certificates of completion and other communication mail proof must be attached	12
5.	Bibliography/References	iii

LIST OF FIGURES

Figure 4.1 High-Level System Architecture

Figure 4.2 Iterative Development Methodology

LIST OF TABLES

Table 4.1 Technologies and Tools Used

Table 4.2 Major Internship Responsibilities

CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

The internship at ThumbbyX Global Private Limited provided practical exposure to the complete software development lifecycle. During the internship, I worked as a Software Developer Intern in the Technology & Product Department and was primarily responsible for the development, deployment, maintenance, debugging, and enhancement of the ThumbbyX web application.

The major project undertaken during the internship was the development of the ThumbbyX website (thumbbyx.com) from the ground up. The project involved developing a functional full-stack web application using the MERN stack and deploying the application on Amazon Web Services (AWS).

The work was not limited to initial development. After developing and deploying the application, I was also involved in maintaining the software, identifying and debugging issues, resolving technical problems, and implementing new features and modifications according to changing requirements.

This internship therefore provided experience in multiple stages of software development, including requirement understanding, application design, frontend development, backend development, database integration, testing, deployment, debugging, maintenance, and feature enhancement.

The project also helped bridge the gap between theoretical knowledge acquired during the B.Tech Computer Science and Engineering program and practical software engineering practices used in a professional environment.

4.2 Organization Profile

ThumbbyX Global Private Limited is a technology-driven startup working in the construction technology. The organization is focused on using technology to make the construction ecosystem more organized, transparent, accessible, and technology-driven. ThumbbyX is building a digital platform that connects construction professionals and creates opportunities for professionals such as contractors, civil engineers, and other construction-related professionals.

The official ThumbbyX website is <https://thumbbyx.com>. The platform has been introduced as a beta product initially targeting construction professionals in locations including **Delhi NCR, Noida, Greater Noida, Gurugram, and Patna**. The platform enables construction professionals to create digital professional profiles, undergo verification, and access relevant project opportunities.

ThumbbyX aims to address the difficulties faced by professionals in the construction industry by providing a centralized digital platform where professionals can build their professional presence and connect with project opportunities. The organization's broader objective is to make the construction ecosystem more efficient and technology-driven.

The company was incorporated as ThumbbyX Global Private Limited in 2025 and is registered as a private limited company in Bihar. Public company records classify its principal activity under computer programming, consultancy and related activities.

During my internship, I worked in the Technology & Product Department as a Software Developer Intern. My primary project was the development and technical improvement of the ThumbbyX web platform. The website was developed from the ground up and subsequently deployed to AWS. My responsibilities also included debugging issues, maintaining the deployed software, and implementing feature upgrades as the requirements of the platform evolved.

The organization can be contacted through its official website and company communication channels. The email address used for ThumbbyX communication is **info@thumbbyx.com**.

Website: <https://thumbbyx.com>

Email: info@thumbbyx.com

4.3 Problem Statement

At the beginning of the internship, an initial version of the ThumbbyX project already existed; however, the application was not in a stable and production-ready state. The backend was not functioning reliably, and the application was experiencing a critical issue in which backend requests could cause the server/application to crash repeatedly. This made the existing system difficult to use and prevented reliable interaction between the frontend and backend.

In addition to the backend stability issues, the frontend was incomplete and required further development to provide the functionality needed by the platform. The existing application therefore required significant development and integration work before it could be deployed and used as a functional web platform.

The major problems identified were:

1. **Unstable backend:** The backend was not handling requests reliably and could crash during requests.
2. **Incomplete frontend:** Several parts of the frontend required development and completion before the application could provide the intended user experience.
3. **Frontend-backend integration issues:** The different parts of the application needed to be properly connected and tested to ensure reliable communication between the client and server.
4. **Deployment requirement:** The application needed to be properly prepared and deployed to a cloud environment so that it could operate as a live website.
5. **Software reliability:** The application required debugging and testing to identify and resolve errors before and after deployment.
6. **Future feature requirements:** Since ThumbbyX was an evolving platform, the software needed to be structured and maintained in a way that allowed new features and improvements to be introduced as requirements changed.

Therefore, the primary problem addressed during the internship was to transform the existing incomplete and unstable application into a functional, stable, deployable, and maintainable web platform.

The project involved rebuilding and completing the required frontend functionality, resolving backend issues, integrating the application components, connecting the required database functionality, deploying the application on AWS, and continuously maintaining and upgrading the software.

The final objective was to make the ThumbbyX platform sufficiently functional and reliable for its intended users while establishing a technical foundation that could be further expanded as the organization continued to develop its product.

4.4 Project Objectives

- To develop the ThumbbyX web application from scratch.
- To build the frontend using React.js and implement the required user-facing functionality.
- To develop backend services and APIs using Node.js and Express.js.
- To integrate the application with MongoDB for persistent data storage.
- To integrate the frontend, backend, and database into a functional full-stack application.
- To deploy the application on Amazon Web Services (AWS).
- To test the application and identify technical issues during development and after deployment.
- To debug and resolve software problems as they occurred.
- To maintain the deployed application and ensure existing functionality continued to work correctly.
- To implement feature upgrades and modifications according to changing project requirements.
- To gain practical experience with the software development lifecycle and production-oriented web development.

4.5 Scope of the Project

The scope of the project covered the development, deployment, maintenance, and continuous enhancement of the ThumbbyX web application.

The project began with development of the website from scratch and extended through full-stack integration and production deployment. After deployment, the scope also included debugging, maintenance, and implementation of additional or modified features as required.

4.5.1 Frontend Development

The frontend was developed using React.js and served as the main user-facing layer of the web application. The work included developing pages, reusable components, forms, navigation, interactive elements, and communication with backend APIs.

The frontend was developed with the objective of providing a functional interface through which users could access the features provided by the platform.

4.5.2 Backend Development

The backend was developed using Node.js and Express.js. It provided server-side application logic and APIs required by the frontend.

Backend responsibilities included processing requests, implementing application logic, communicating with the database, and returning appropriate responses to the frontend.

4.5.3 Database Integration

MongoDB was used for persistent data storage. The backend communicated with the database to create, retrieve, update, and manage application-related information.

4.5.4 AWS Deployment

The completed web application was deployed on AWS. The deployment stage involved preparing the application for a production environment, configuring the required cloud environment, deploying the application, and verifying that the website operated correctly after deployment.

4.5.5 Maintenance and Feature Upgrades

The project scope continued after deployment. Software issues were investigated and debugged whenever they occurred, and fixes were tested before being considered complete.

The application was also upgraded when new requirements or feature changes were communicated. This required modifying existing functionality and adding new functionality while maintaining the stability of the existing application.

4.6 Technologies and Tools Used

The primary technology stack used during the internship was the MERN stack, supported by AWS for cloud deployment.

Technology / Tool	Purpose	Application in Project
MongoDB	Database	Persistent storage and management of application data.
Express.js	Backend framework	API development, routing, request handling, and server-side logic.
React.js	Frontend library	Development of the interactive user interface.
Node.js	Runtime environment	Execution of backend JavaScript services.

MERN Stack	Full-stack architecture	Integration of frontend, backend, and database technologies.
AWS	Cloud platform	Deployment and hosting of the web application.

Table 4.1 Technologies and Tools Used

4.6.1 MongoDB

MongoDB was used as the database technology for storing application-related information. It provided the persistent data layer used by the backend services.

4.6.2 Express.js

Express.js was used as the backend web application framework. It supported server-side routing, API development, request processing, and communication between the application logic and database.

4.6.3 React.js

React.js was used to develop the frontend of the ThumbbyX web application. It supported component-based interface development and interaction with backend services.

4.6.4 Node.js

Node.js was used as the server-side JavaScript runtime for the backend of the application.

4.6.5 Amazon Web Services (AWS)

AWS was used for cloud deployment of the ThumbbyX application. Deployment moved the application from the development environment to a production-accessible environment.

4.7 System Architecture

The ThumbbyX application follows a client-server based full-stack architecture using the MERN stack. The major application layers are the user interface, backend/API layer, and database layer, with the application deployed through an AWS-based cloud environment.

The user interacts with the application through a web browser. The React frontend handles the user interface and sends requests to the backend. The Node.js and Express.js

backend processes those requests and communicates with MongoDB for persistent data operations. The deployed application is hosted in the AWS environment.

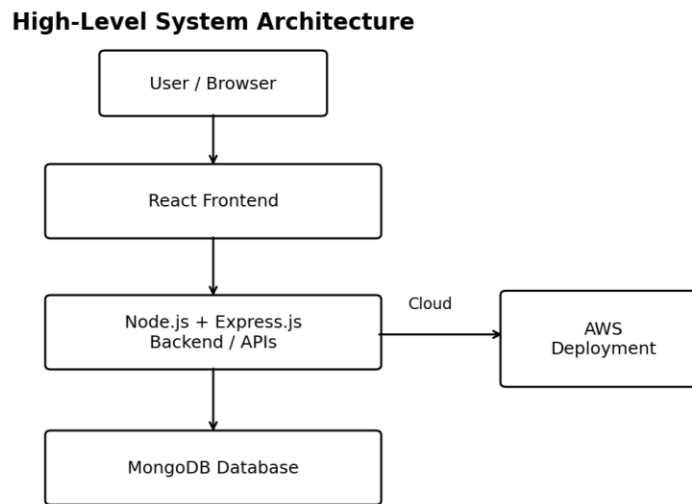


Figure 4.1 High-Level System Architecture

4.7.1 Frontend Layer

The frontend is the user-facing portion of the application. React.js components are used to create the interface and provide interaction with the platform. User actions generate requests that are communicated to the backend through APIs.

4.7.2 Backend Layer

The backend provides server-side application logic. Node.js and Express.js are used to process requests, implement application operations, communicate with the database, and return responses to the frontend.

4.7.3 Database Layer

MongoDB acts as the persistent data layer. The backend communicates with MongoDB to store and retrieve the data required by the application.

4.7.4 Deployment Layer

AWS provides the cloud environment in which the application is deployed. The deployment layer allows the completed application to be accessed through the ThumbbyX website.

4.8 Methodology

The project followed an iterative software development approach. Since the application was developed from scratch and subsequently maintained and upgraded, development

progressed through repeated stages of implementation, testing, debugging, deployment, and enhancement.

Iterative Development Methodology



Figure 4.2 Iterative Development Methodology

- 1. Requirement Understanding:** The first stage involved understanding the purpose of the ThumbbyX platform and identifying the functionality required for the website.
- 2. Application Planning:** The application structure and technical approach were planned using the MERN stack and AWS for deployment.
- 3. Frontend Development:** The frontend was developed using React.js, including pages, components, forms, navigation, and interactive functionality.
- 4. Backend Development:** Backend services and APIs were developed using Node.js and Express.js to implement server-side functionality.
- 5. Database Integration:** MongoDB was integrated with the backend to provide persistent data storage and retrieval.
- 6. Frontend-Backend Integration:** The React frontend was connected with backend APIs so that user actions could communicate with server-side functionality.
- 7. Testing and Debugging:** The application was tested during development. Issues were reproduced, analyzed, fixed, and retested.
- 8. AWS Deployment:** The application was prepared for production and deployed on AWS. The deployed version was then verified.
- 9. Maintenance:** After deployment, software issues were investigated and resolved, and the application was maintained.
- 10. Feature Enhancement:** Existing functionality was modified and new features were implemented as project requirements evolved.

4.8.1 Major Internship Responsibilities

Area	Responsibility
Web Development	Developed the ThumbbyX website from scratch using the MERN stack.

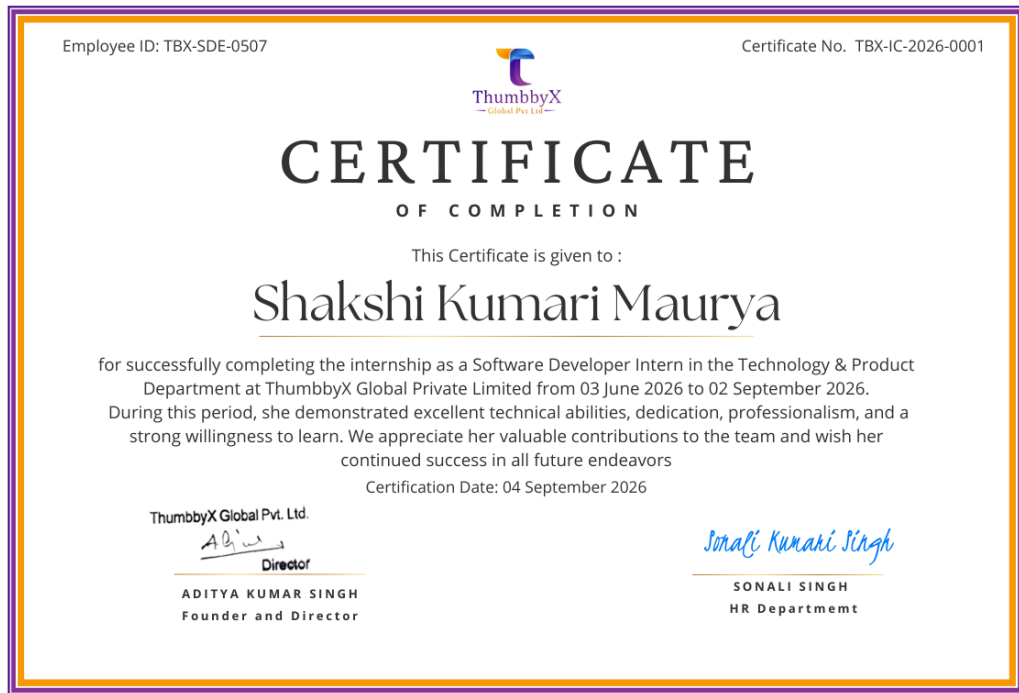
Frontend	Implemented user-facing interfaces and interactive functionality using React.js.
Backend	Developed server-side functionality and APIs using Node.js and Express.js.
Database	Integrated MongoDB for persistent application data.
Deployment	Deployed the application on AWS.
Debugging	Investigated and fixed software issues and errors.
Maintenance	Maintained the deployed software and verified existing functionality.
Feature Upgrades	Implemented new features and modified existing functionality according to requirements.

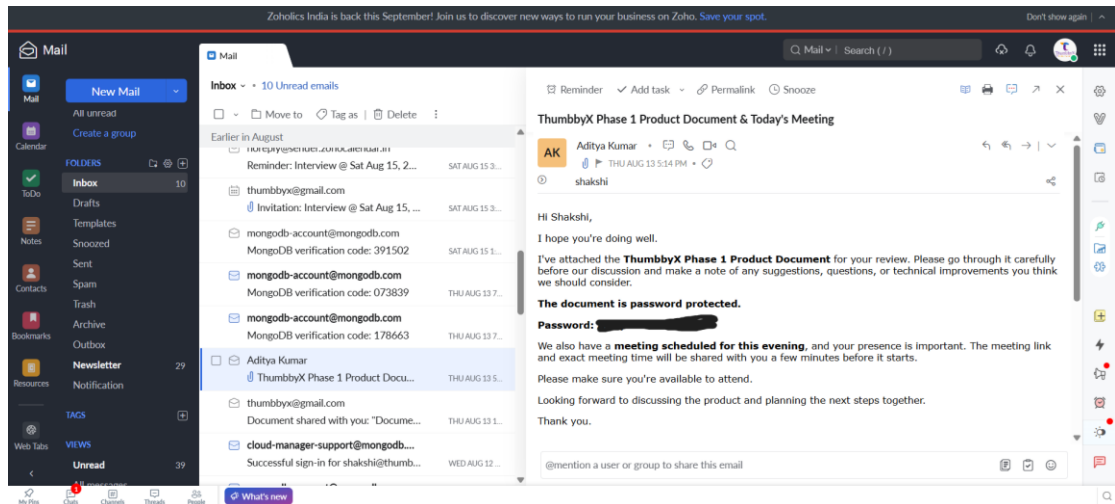
Table 4.2 Major Internship Responsibilities

4.9 Expected Outcomes

- A fully functional ThumbbyX web application developed from scratch.
- Successful integration of frontend, backend, and database components using the MERN stack.
- Successful deployment of the application on AWS.
- A maintainable software system that can be debugged and modified as requirements change.
- Continuous improvement of the application through feature upgrades.
- Practical experience in full-stack development and cloud deployment.
- Improved understanding of debugging, maintenance, testing, and production-oriented software development.

4.10 Certificates and Communication / Mail Proof





CHAPTER 5

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